

COALGUARD SMART GOVERNANCE PLATFORM

COMPLETE PROJECT AUDIT & FEATURE INVENTORY REPORT

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1. PROJECT OVERVIEW

- Project Name:** CoalGuard (Coal Mine Smart Governance & Compliance System)
- Application Purpose:** Smart governance, statutory compliance monitoring, hazard reporting, visual camera AI surveillance, and emergency distress management for underground and opencast coal mines. Designed under DGMS (Directorate General of Mines Safety) guidelines, Coal Mines Regulations (CMR) 2017, and the Mines Act 1952.
- Main Use Cases:**
 - Real-time underground miner safety reporting via voice NLP, hand gestures (MineSign AI), and SOS distress beacons.
 - Shift supervisor operational monitoring, CCTV video surveillance (people counting, helmet compliance, smoke/fire detection), field defect management, and hazard triage.
 - National regulatory authority statutory oversight, 2-step violation closure verification, cross-mine risk analytics, and immutable SHA-256 cryptographic audit tracking.
- Technology Stack:**
 - Frontend:** React 19 (v19.2.8), Vite 8 (v8.3.0), Tailwind CSS v4, Lucide React (v1.45.0), Recharts (v3.10.1), Leaflet & React-Leaflet (v5.0.0), Tesseract.js (v7.0.0 for OCR).
 - Backend:** Node.js ESM Express server (v4.21.2), Node-Cron, PDFKit, BcryptJS, JSONWebToken.
 - Database / Storage:** Dual-tier database architecture. PostgreSQL (v8.13.1) pool connection if `DATABASE\_URL` is set, with automatic fallback to an in-memory embedded state engine (mockStore.js) persisted with file-backed logs. Client side uses `localStorage` for sessions, offline field reports, and read-notification timestamps.
 - AI / ML Stack (Python):** Python 3.12/3.13, PyTorch, Ultralytics YOLOv8 / YOLOWorld v2, MediaPipe Holistic Landmarker (holistic_landmarker.task), OpenCV (cv2), Scikit-Learn (joblib), Flask / FastAPI HTTP server on port 5005 for live gesture inference.
 - Authentication System:** JWT Bearer token authentication via /api/auth/login and /api/auth/quick-login/:role. Roles: `miner`, `supervisor`, `authority` (with aliases `corporate` and `regulator`).
 - State Management:** React Context (AuthContext.jsx) managing user session, token, language state (English/Hindi), offline report sync queue, and global unread notifications.
 - Routing System:** Custom role-based conditional rendering in App.jsx (switch(user.role)) and state-driven sub-tab navigation stacks with back history (navHistory) within each portal.
 - Styling / UI Libraries:** Tailwind CSS v4, custom government theme styles (gov-panel, gov-table, badge-gov, btn-gov-primary), Lucide React icons, Recharts for analytics.
 - Map / GIS Libraries:** Leaflet v1.9.4 and React-Leaflet v5.0.0 with CARTO OpenStreetMap / Esri satellite tile layers.
 - Build / Run System:** Vite (npm run dev in client), Node.js (npm start in server), Python virtual environment (minesign_ai/.venv).

Key Package Dependencies & Purpose

Package	Location	Purpose
react, react-dom	client/package.json	Core UI library (v19.2.8)
vite	client/package.json	High-speed frontend build tool & dev server
@tailwindcss/vite, tailwindcss	client/package.json	Utility-first CSS framework (v4.3.3)
leaflet, react-leaflet	client/package.json	GIS mapping library for mine boundary polygons and GPS markers
recharts	client/package.json	Charting library for production vs target bar charts and risk trends
tesseract.js	client/package.json	Client-side OCR engine for scanning paper statutory notices
lucide-react	client/package.json	Icon set for UI controls and statutory badges
express	server/package.json	Web application framework for REST API
pg	server/package.json	PostgreSQL database client driver
jsonwebtoken	server/package.json	JWT token sign and verification
bcryptjs	server/package.json	Password hashing
node-cron	server/package.json	Automated statutory escalation background scheduler
pdfkit	server/package.json	Dynamic PDF generation for statutory reports

2. COMPLETE FOLDER / FILE STRUCTURE

```
Coal mines copy/
% % % client/                # Frontend Vite React Application
% % % % public/              # Static video assets & icons
% % % % % videos/            # Demo CCTV streams (people_counter_demo.mp4, helmet_detection_demo.mp4,
smoke_detection_demo.mp4)
% % % % % src/
% % % % % % assets/          # Static image assets (hero.png)
% % % % % % % components/    # AI Analytics components (Model4RiskScoringView, Model5ClassifierView,
% % % % % % % % analytics/   Model6WhatIfSimulatorView)
% % % % % % % % % assistant/  # Floating AI voice/chat assistant (FloatingAssistant.jsx)
% % % % % % % % % authority/  # Authority portal views (AuthorityAIAssessmentsView,
AuthorityDailyGovernanceSummary)
% % % % % % % % % common/    # Shared components (GovHeader, GovNoticeTicker, GovBreadcrumbs,
EmergencySOSBanner, SupervisorDailyLogView, OCRScanModal, TwoStepViolationModal, VoiceHazardModal,
MiningIcons)
% % % % % % % % % maps/      # GIS Map component (MineGISMap.jsx)
% % % % % % % % % minesign/  # Gestural AI components (MineSignView.jsx, MinerCameraModal.jsx)
% % % % % % % % % surveillance/ # CCTV AI surveillance feed viewer (CCTVSurveillanceView.jsx)
% % % % % % % % % context/   # AuthContext.jsx (Authentication, Notifications, Offline Sync)
% % % % % % % % % i18n/      # Dual-language translations (translations.js)
% % % % % % % % % pages/
% % % % % % % % % % Login.jsx # Credentials & Quick-Login portal
% % % % % % % % % % miner/    # MinerDashboard.jsx
% % % % % % % % % % supervisor/ # SupervisorDashboard.jsx
% % % % % % % % % % authority/ # AuthorityDashboard.jsx
% % % % % % % % % % App.jsx    # Role-based root router
% % % % % % % % % % index.css  # Custom government styling & Tailwind directives
% % % % % % % % % % main.jsx   # React entry point
% % % % % % % % % % package.json
% % % % % % % % % % server/    # Backend Node.js Express Application
% % % % % % % % % % % db/      # Database pool, schema.sql, and fallback mockStore.js
% % % % % % % % % % % middleware/ # Authentication middleware (auth.js)
% % % % % % % % % % % routes/  # Express REST routes (auth, compliance, inspections, violations,
analytics, fieldReports, assistant, gis, contractors, workflow, auditTrail, grievances, attendance,
emergency, mlModels, minesign, communication, operationalLog)
% % % % % % % % % % % % utils/ # mlRunner.js, ml_bridge.py, riskEngine.js, hashChain.js, verification
scripts
% % % % % % % % % % % % index.js # Server entry point & cron scheduler
% % % % % % % % % % % % package.json
% % % % % % % % % % % % minesign_ai/
% % % % % % % % % % % % % temporal_engine.py # Standalone Python MineSign Gesture AI Pipeline
% % % % % % % % % % % % % % feature_extractor.py # Sliding window state machine & gesture classifier
% % % % % % % % % % % % % % inference_server.py # MediaPipe landmark normalizer
% % % % % % % % % % % % % % % models/          # Flask API server running on port 5005
% % % % % % % % % % % % % % % % % % % % % % % # Trained classifier joblib models & MediaPipe task file
% % % % % % % % % % % % % % % % % % % % % % % # YOLOv8 / YOLOWorld PPE Helmet Detection model & dataset
% % % % % % % % % % % % % % % % % % % % % % % # People ingress/egress counter script with line crossing tracker
% % % % % % % % % % % % % % % % % % % % % % % # Smoke & fire chromatic video analyzer script
% % % % % % % % % % % % % % % % % % % % % % % # MSHA accident dataset tabular model & CSV dataset
% % % % % % % % % % % % % % % % % % % % % % % # Text hazard NLP classifier training pipeline
% % % % % % % % % % % % % % % % % % % % % % % # What-If risk trajectory simulator dataset
```

3. ALL PORTALS

A. MINER PORTAL

- **Login / Entry Flow**: User logs in with `role: 'miner'`. Handled by `App.jsx` which renders `MinerDashboard.jsx`.
- **Home / Dashboard**: Overview cards showing Active Shift status, Emergency SOS Banner, Quick Action Grid (Voice Hazard, Camera Gestures, Ask Supervisor, Biometric Check-in), and My Recent Field Submissions table.
- **Navigation Tabs**:
 - `Overview & Shift Check-In` (`home`)
 - `MineSign Camera Gesture Reporting` (`gestures`)
 - `My Submitted Hazard Reports` (`my\_reports`)
 - `Direct Inquiries to Supervisor` (`ask\_supervisor`)
 - `Statutory Safety Circulars` (`safety`)
 - `Official Notifications Feed` (`notifications`)
 - `Worker Profile & DGMS Competency` (`profile`)
 - `Colliery Life-Safety Distress Beacon` (`sos`)
- **Key Components**: `GovHeader`, `GovNoticeTicker`, `GovBreadcrumbs`, `VoiceHazardModal`, `OCRScanModal`, `MinerCameraModal`, `FloatingAssistant`.
- **Notification Mechanism**: Polls `/api/workflow/notifications` every 15 seconds. Unread red dots appear on `my\_reports`, `ask\_supervisor`, `notifications`, and `sos`. Disappear when section tab is visited (persisted in `localStorage.coalguard\_miner\_seen\_sections`).

B. SUPERVISOR PORTAL

- ****Login / Entry Flow****: User logs in with `role: 'supervisor'`. Rendered via `SupervisorDashboard.jsx`.
- ****Home / Dashboard****: Control Strip (Mine/Sector info), KPI Summary Strip (Open Issues, Pending Verification, Active Emergencies, Compliance Items, Worker Requests, Escalated Cases), Pending Supervisor Action Items table, Active Emergency Alerts table, Central Daily Operational Log (`SupervisorDailyLogView`), and CCTV Surveillance Quick Access strip.
- ****Navigation Tabs****:
 - `Operational Control Center` (`overview`)
 - `CCTV Surveillance` (`cameras`) [Sub-tabs: Counter, Helmet, Smoke]
 - `MineSign Gestural Verification` (`gestures`)
 - `AI Predictive Models` (`models`) [Sub-tabs: Risk Scoring, What-If Simulator]
 - `Active Emergency Incidents` (`emergencies`)
 - `Field Reports & Defect Log` (`field\_reports`)
 - `Statutory Compliance Register` (`compliance`)
 - `Zone Geographic GIS Map` (`map`)
 - `Communication Channels` (`communication`)
 - `Official Notices & Gazettes` (`updates`)
- ****Key Components****: `CCTVSurveillanceView`, `MineSignView`, `Model4RiskScoringView`, `Model6WhatIfSimulatorView`, `SupervisorDailyLogView`, `MineGISMap`, `OCRScanModal`.

C. AUTHORITY PORTAL

- ****Login / Entry Flow****: User logs in with `role: 'authority'`, `corporate`, or `regulator`. Rendered via `AuthorityDashboard.jsx`.
- ****Home / Dashboard****: Scope Selector (All Mines vs Specific Mine), Executive Top Banner, 5 Metric Cards (Statutory Compliance %, Open Violations, Shift Attendance, AI Assessments Received, DGMS SHA-256 Audit Chain), Cross-Mine Risk Index Table, Urgent Action Queue, and Central Consolidated Daily Governance Summary.
- ****Navigation Tabs****:
 - `National Governance Overview` (`overview`)
 - `Demarcated Mine Sectors & Workforce` (`mines`)
 - `Statutory Compliance & Two-Step Violations` (`compliance` / `inspections\_violations`)
 - `Escalated Colliery Emergencies` (`emergencies`)
 - `Supervisor AI Assessments Received` (`ai\_assessments`)
 - `Cross-Sector Risk Analytics` (`analytics`)
 - `National GIS Geospatial Layer` (`map`)
 - `Supervisor Communications & Grievances` (`communication` / `reports`)
 - `Immutable SHA-256 Cryptographic Audit Ledger` (`audit`)
 - `DGMS Gazettes & Circular Bulletins` (`updates`)
- ****Key Components****: `AuthorityAIAssessmentsView`, `AuthorityDailyGovernanceSummary`, `TwoStepViolationModal`, `MineGISMap`, `OCRScanModal`.

4. MINER PORTAL — COMPLETE FEATURE INVENTORY

- ****Shift Biometric Attendance Check-In****
- ****Purpose****: Verify shift entry for Form B Muster Roll.
- ****User Trigger****: Clicks "Confirm Attendance" button on Home tab.
- ****Data Flow****: `POST /api/attendance/check-in` -> saved in database/mockStore -> updates Supervisor Attendance summary.
- ****Voice Hazard Report****
- ****Purpose****: Record spoken hazard observations in Hindi/English with AI NLP auto-categorization.
- ****User Trigger****: Clicks "Voice Hazard Report" button -> opens `VoiceHazardModal`.
- ****Data Flow****: Web Speech API speech-to-text -> `POST /api/models/classify-hazard` (NLP categorizes category & severity) -> `POST /api/field-reports` -> routed directly to Supervisor's Field Reports list.
- ****Manual Field Safety Report****
- ****Purpose****: Submit written hazard reports with category, severity, and description.
- ****User Trigger****: Clicks "Create Safety Report" -> opens modal form -> submits.
- ****Data Flow****: `POST /api/field-reports` -> saved in `mockStore.field\_reports` -> visible to Supervisor.
- ****Camera Gestures (MineSign AI)****
- ****Purpose****: Transmit silent visual hand signals to alert Supervisor via laptop webcam or direct trigger.

- **User Trigger**: Opens MineSign Camera Modal or clicks "Transmit Signal" button under Gestures catalog.
 - **Data Flow**: Client webcam -> MediaPipe landmarks -> `POST /api/minesign/events` (creates candidate event `PENDING\_CONFIRMATION` with 5s duplicate cooldown) -> appears on Supervisor MineSign tab awaiting visual confirmation.
 - **Colliery Life-Safety Distress Beacon (SOS)**
 - **Purpose**: Instantly transmit GPS coordinates and high-priority alarm during emergencies.
 - **User Trigger**: Clicks "TRIGGER EMERGENCY SOS" button.
 - **Data Flow**: Geolocation API captures coordinates -> `POST /api/emergency/sos` -> creates active SOS incident in `mockStore.sos\_events` -> triggers emergency sound and banner on Supervisor Dashboard.
 - **Ask Supervisor (Direct Inquiries)**
 - **Purpose**: Submit questions or equipment requests directly to the shift in-charge.
 - **User Trigger**: Fills out form under "Ask Supervisor" tab -> submits.
 - **Data Flow**: `POST /api/communication/miner-queries` -> stored in `mockStore.miner\_queries` -> appears under Supervisor Communication tab for reply.
 - **Statutory Safety Circulars**
 - **Purpose**: View DGMS advisories and colliery bulletins (UI rendering static statutory guidelines).
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5. SUPERVISOR PORTAL — COMPLETE FEATURE INVENTORY

- **CCTV Surveillance System (3 Video AI Models)**
 - **Features**:
 - **People Counter**: Simulates shaft entrance camera (`CAM-Z3-01`), tracks ingress (+1), egress (-1), and net underground occupancy (6 / 45 max threshold). Syncs cumulative count via `POST /api/models/cctv/people-counter-sync`. "Sync Muster Headcount" button broadcasts headcount via `api/models/cctv/people-headcount`.
 - **Safety Helmet Detector**: Scans checkpoint video (`CAM-CHK-02`) using YOLOWorld v2. Detects missing or damaged helmets. Supervisor can click "Dismiss" (logs audit) or "Raise Field Issue" (`POST /api/models/cctv/helmet-action`), creating an open Field Report assigned to Shift Safety Sirdar.
 - **Smoke & Fire Vision**: Scans haulage road video (`CAM-HAUL-03`). Upon detection, Supervisor selects action: Field Report, Authority Escalation, or Both (`POST /api/models/cctv/smoke-action`).
 - **MineSign Gestural Confirmation**
 - **Purpose**: Review incoming visual gesture candidate events (`PENDING\_CONFIRMATION`).
 - **User Trigger**: Supervisor clicks "Confirm & Create Event" or "Reject".
 - **Data Flow**: Confirming (`PATCH /api/minesign/events/:id/confirm`) converts state to `OPEN`, creates official safety notification, and writes SHA-256 audit entry. Rejection (`PATCH /api/minesign/events/:id/reject`) marks `REJECTED` without creating open alerts. Supervisor can also route confirmed events to Field Reports or Authority.
 - **AI Models: Mine Risk Scoring & What-If Simulator**
 - **Features**:
 - **Model 4 (Risk Scoring)**: Inputs equipment age, methane level, shift type, inspection gap -> calculates risk percentage via `POST /api/models/predict-risk`.
 - **Model 6 (What-If Simulator)**: Interactive sliders for rainfall, worker experience, equipment age, and shift hours -> simulates risk trajectory via `POST /api/models/simulate-whatif`.
 - **Send Assessment to Authority**: Clicks button -> `POST /api/models/send-assessment`. Creates immutable snapshot of selected inputs and output, stored in `mockStore.ai\_assessments`. Modifying inputs later in the simulator does NOT alter previously transmitted assessment records.
 - **Field Reports Management**
 - **Purpose**: Triage worker reports. Supervisor inspects details modal, acknowledges (`POST /api/field-reports/:id/ack`), or closes issue with mandatory resolution notes (`PATCH /api/field-reports/:id/status` -> `CLOSED`).
 - **Emergency Incident Handling**
 - **Purpose**: Manage active SOS distress beacons. Supervisor acknowledges (`POST /api/emergency/sos/:id/ack`), resolves (`POST /api/emergency/sos/:id/resolve`), or escalates to Authority (`POST /api/emergency/sos/:id/escalate`).
 - **Communication Gateway**
 - **Features**: Reply to miner inquiries (`PATCH /api/communication/miner-queries/:id/respond`). Submit communication requests to Authority (`POST /api/communication/requests`). Once approved by Authority, open live chat thread (`POST /api/communication/messages`).
 - **GIS Spatial Map**
 - **Features**: Renders React-Leaflet map with zone polygons, active incident markers, and camera locations.
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6. AUTHORITY PORTAL — COMPLETE FEATURE INVENTORY

- ****National Governance & Mine Scope Filtering****
- ****Purpose****: Filter all datasets across all mines (`all`) or specific mine IDs (`mine-demo-01`, `mine-raniganj-02`, etc.).
- ****Statutory Compliance & Two-Step Violation Closures****
- ****Purpose****: Enforce DGMS strict 2-step violation closure.
- ****User Trigger****: Authority clicks "Verify & Sign-off" -> opens `TwoStepViolationModal`.
- ****Data Flow****:
 - Step 1 (Supervisor/Field): Upload photo proof (`PATCH /api/violations/:id/step1`). State becomes `rectification\_submitted`.
 - Step 2 (Authority Only): Authority reviews photo proof, selects "Approve & Formally Close Violation" (`PATCH /api/violations/:id/step2`), transitioning state to `verified\_closed` and generating SHA-256 audit entry.
- ****Supervisor AI Assessments Received****
- ****Purpose****: Review assessments sent by Supervisors (`AuthorityAIAssessmentsView.jsx`).
- ****Actions****: View exact input/output snapshot, inspect key risk drivers, and update status from `NEW` -> `REVIEWED` or `ACKNOWLEDGED` (`PATCH /api/models/assessments/:id/status`).
- ****Escalated Colliery Emergencies Queue****
- ****Purpose****: Authority receives ONLY emergencies escalated by Supervisors or critical SOS events.
- ****Actions****: Acknowledge escalation (`POST /api/emergency/sos/:id/ack`) or mark case resolved & closed (`POST /api/emergency/sos/:id/resolve`).
- ****Communication Approval & Direct Dispatch****
- ****Features****:
 - Review supervisor communication requests (`PATCH /api/communication/requests/:id/decision` -> `approved` / `rejected`).
 - Directly initiate statutory directives to supervisors without prior request (`POST /api/communication/requests` + auto-approve + initial message).
- ****Cryptographic SHA-256 Audit Trail****
- ****Purpose****: Inspect immutable blockchain-lite audit ledger.
- ****Actions****: "Verify Cryptographic Integrity" (`POST /api/audit/verify-integrity`), "Simulate Tamper" (`POST /api/audit/simulate-tampering`), and "Restore" (`POST /api/audit/restore-integrity`).
- ****Statutory PDF Report Generator****
- ****Purpose****: Generate downloadable PDF dossier via `POST /api/workflow/generate-statutory-report/:mineId` using PDFKit.

7. CROSS-PORTAL DATA FLOW

[MINER PORTAL]	
% % % Voice / Safety Report % % % % > POST /api/field-reports % % % % % % % % % % > [SUPERVISOR]	
% % % Ask Supervisor Query % % % % % > POST /api/communication/miner-queries % % % % > (Inspects, Verifies &	
% % % Camera Gesture Signal % % % % % > POST /api/minesign/events (PENDING) % % % % % % % % > (Confirms Gestur	
% % % Life-Safety SOS Beacon % % % % > POST /api/emergency/sos % % % % % % % % % % % % % % % % > (Acknowledge	
	% (Escalated)
SOS / Communication / AI Snapshot / 2-Step Proof)	%%
	[AUTHORITY PORTAL] (Reviews AI Snapshot,
Approves Comm Channel,	
	Executes Step-2
Physical Sign-off,	
	Verifies SHA-256 Hash
Chain)	

8. NOTIFICATION SYSTEM

- **Generation**: Triggered by backend API handlers when field reports, SOS events, MineSign confirmations, or AI assessments are submitted.
- **Storage**: Appended to ``mockStore.notifications`` in server memory. Fetched by client ``AuthContext`` via ``GET /api/workflow/notifications`` every 15 seconds.
- **Red Dots**: Section-specific indicators. Checked by comparing latest item timestamp against ``seenSections`` timestamp saved in ``localStorage``:
 - ``coalguard_miner_seen_sections``
 - ``coalguard_supervisor_seen_sections``
 - ``coalguard_authority_seen_sections``
- **Behavior**: When a user clicks a tab, ``handleSelectTab`` records ``Date.now()`` into ``seenSections``, causing the red dot for that section to disappear immediately.

- **Numeric Badges vs Red Dots**: Numeric badges represent live dataset counts (e.g. `6 / 45 max`), while red dots indicate unseen/unread items.
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9. NAVIGATION / ROUTES

- **Client Router Architecture**: `App.jsx` uses role-based switch statement without React Router URL paths (`switch(user.role)`). Page views are managed via React `activeTab` / `activeSection` state inside `MinerDashboard.jsx`, `SupervisorDashboard.jsx`, and `AuthorityDashboard.jsx`.
 - **Breadcrumbs & Back Navigation**: `GovBreadcrumbs.jsx` tracks `navHistory` stack array.
 - "Go Back" (`onBack`): Pops previous tab from `navHistory` stack or closes open modal/detail view.
 - "Go Home" (`onHome`): Resets `navHistory` to empty array and sets active tab to `home` / `overview`.
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10. AI / CAMERA SYSTEM

- **Detection Models & Libraries**:
 - **People Counter**: OpenCV + YOLOv8n + ByteTrack centroid tracker (`people counter/people\_exit\_counter/main.py`). Virtual line crossing detection at Y=350. Client UI plays pre-annotated simulation feed (`/videos/people\_counter\_demo.mp4`).
 - **Safety Helmet Detector**: YOLOWorld v2 (`yolov8s-worldv2.pt` / CLIP ViT-B-32). Detects missing or damaged helmets (`helmet detection/helmet\_detection/inference/run\_inference.py`). Client UI plays pre-annotated simulation feed (`/videos/helmet\_detection\_demo.mp4`).
 - **Smoke & Fire Vision**: HSV chromatic flame segmentation + MOG2 motion subtractor (`SMOKE DETECTION/analyze\_video.py`). Client UI plays pre-annotated simulation feed (`/videos/smoke\_detection\_demo.mp4`).
 - **MineSign Gesture AI**: Flask server running `inference\_server.py` on port 5005. MediaPipe Holistic Landmarker (`holistic\_landmarker.task`) extracts 126 normalized hand/body keypoints -> 30-frame sliding window temporal engine (`temporal\_engine.py`) -> Scikit-learn classifier (`minesign\_classifier\_v2.joblib`).
 - **Simulated vs Real**: Real Python inference scripts exist in repository. Web UI displays synchronized video streams with live API interaction buttons to trigger backend event pipelines.
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11. AI MODEL FEATURES

Model 4: Explainable Mine Risk Scoring

- **Inputs**: Equipment Age, Methane Level %, Shift Type, Days Since Inspection, Recent Incident Count, Humidity %.
- **Calculation**: Weighted risk scoring algorithm (`utils/riskEngine.js` / `utils/mlRunner.js`).
- **Outputs**: Predicted Risk %, Risk Level (`LOW`, `WATCH`, `HIGH`, `CRITICAL`), Key Drivers, Trajectory.

Model 6: What-If Risk Trajectory Simulator

- **Inputs**: Rainfall Intensity (mm/hr), Worker Average Experience (yrs), Equipment Maintenance Score %, Shift Operating Hours.
- **Outputs**: Simulated Risk Score, Baseline Comparison, Trajectory, Recommended Mitigation Directives.

"Send Assessment to Authority" Workflow

- **Exact Behavior**: Creates a deep-clone snapshot (`JSON.parse(JSON.stringify(inputs))` and `output`), assigns a unique ID (`RSA-xxxx` / `WIA-xxxx`), and stores it in `mockStore.ai\_assessments`.
 - **Immutability**: Changing input sliders later in the simulator does **NOT** modify previously sent assessment records.
 - **Authority Handling**: Authority receives assessment in `AuthorityAIAssessmentsView.jsx`, views exact snapshot, and updates status (`NEW` -> `REVIEWED` / `ACKNOWLEDGED`).
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12. MAP / GIS

- **Library**: Leaflet v1.9.4 & React-Leaflet v5.0.0 (`components/maps/MineGISMap.jsx`).
 - **Tile Providers**: CARTO OpenStreetMap / Esri World Imagery (No paid API key required).
 - **Configuration & Coordinates**: Centered at Jharia Coalfield, Jharkhand (Lat: `23.7508`, Lng: `86.4192`).
 - **Features**: Interactive mine boundary polygons (Zone 1 to Zone 5), camera node markers, and emergency GPS distress pins.
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13. DATA STORAGE

Entity / Event | Creation Point | Storage Mechanism | Shared Across Portals? | Persistence
User Session & Auth Token | `Login.jsx` | `localStorage` (`coalguard\_user`, `coalguard\_token`) | Yes | Persists across refresh & restart
Field Reports | `MinerDashboard.jsx` / CCTV | `mockStore.field\_reports` / Postgres | Miner -> Supervisor -> Authority | In-memory / Postgres DB
Emergency SOS Events | `MinerDashboard.jsx` | `mockStore.sos\_events` / Postgres | Miner -> Supervisor -> Authority | In-memory / Postgres DB
MineSign Gesture Candidates | `MinerCameraModal.jsx` | `mockStore.minesign\_events` | Miner -> Supervisor | In-memory
AI Assessment Snapshots | `Model4RiskScoringView` | `mockStore.ai\_assessments` | Supervisor -> Authority | In-memory
Violations & 2-Step Proof | `AuthorityDashboard.jsx` | `mockStore.violations` / Postgres | Supervisor <-> Authority | In-memory / Postgres DB
Cryptographic Audit Ledger | Backend API handlers | `mockStore.audit\_logs` / Postgres | Authority | In-memory / File-backed
Offline Reports Queue | `AuthContext.jsx` | `localStorage` (`coalguard\_offline\_reports`) | Miner (Local) | Persists until online sync

14. FORMS AND USER INPUTS

- **Voice / Manual Hazard Report Form**: Input category, severity (`low`, `medium`, `high`, `critical`), description, zone. Submits to `/api/field-reports`.
 - **Supervisor Comm Request Form**: Input topic, urgency, description, zone. Submits to `/api/communication/requests`.
 - **Authority Violation 2-Step Form**: Step 1 (Photo proof URL & remarks) -> Step 2 (Physical sign-off confirmation & form B signature).
 - **Contractor Onboarding Form**: Input agency name, reg number, service type, crew count, PESO license number. Submits to `/api/contractors`.
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15. TABLES / LISTS / FILTERS

- **Supervisor Action Items Table**: Filter by severity/status. Actions: View details, Verify on-site, Close issue.
 - **Authority Cross-Mine Risk Table**: Columns for Mine Name, Subsidiary, Risk Score, Compliance %, Active Violations. Filtered by Mine Scope dropdown.
 - **SHA-256 Audit Trail Table**: Columns for Block #, Timestamp, Action & Entity, Actor, SHA-256 Hash. Includes text search filter across hashes and actors.
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16. MODALS / POPUPS / DIALOGS

- **TwoStepViolationModal.jsx**: Triggered from Authority/Supervisor violation tables. Displays Step 1 proof and Step 2 sign-off controls.
 - **VoiceHazardModal.jsx**: Triggered from Miner Home tab. Captures browser audio, displays live transcript, and calls NLP classifier.
 - **OCRScanModal.jsx**: Triggered from header ribbon. Uploads or scans document images using Tesseract.js to extract text into compliance forms.
 - **MinerCameraModal.jsx**: Triggered from Miner Gestures tab. Displays webcam feed and MineSign AI real-time gesture telemetry.
-

17. COMMUNICATION SYSTEM

- **Miner !" Supervisor**: Miner submits queries (`POST /api/communication/miner-queries`). Supervisor views and replies (`PATCH /api/communication/miner-queries/:id/respond`).
 - **Supervisor !" Authority (Request Approval Flow)**: Supervisor submits topic request (`POST /api/communication/requests`). Status is `pending`. Authority approves or rejects (`PATCH /api/communication/requests/:id/decision`). Once approved, live chat messages can be posted (`POST /api/communication/messages`).
 - **Authority !" Supervisor (Direct Directive)**: Authority can directly initiate an auto-approved communication thread without prior request.
-

18. EMERGENCY SYSTEM

- **Trigger**: Miner clicks "TRIGGER EMERGENCY SOS" -> `POST /api/emergency/sos``.
- **Supervisor Actions**: Alarm sounds on Supervisor dashboard. Supervisor can Acknowledge (`/ack``), Resolve (`/resolve``), or Escalate (`/escalate``).
- **Authority Actions**: Authority receives escalated SOS events. Authority can Acknowledge or Resolve & Close case.

19. AUDIT / LOGGING

- **Mechanism**: Every critical state change calls `appendAuditLog()` (`server/db/mockStore.js``).
- **Cryptographic Hashing**: Computes SHA-256 hash chaining `current_hash = SHA256(previous_hash + block_number + timestamp + action + payload)``.
- **Integrity Verification**: `POST /api/audit/verify-integrity`` re-computes hashes sequentially. "Simulate Tamper" intentionally corrupts a payload to demonstrate break detection; "Restore" fixes the chain.
- **Daily Operational Log**: `SupervisorDailyLogView`` and `AuthorityDailyGovernanceSummary`` render a consolidated event timeline.

20. CURRENT HOME/DASHBOARD CONTENT

- **Supervisor Home**: Control Strip, 6 KPI cards, Pending Action Items table, Active Emergencies table, Central Daily Operational Log timeline (`SupervisorDailyLogView``), and CCTV quick-access strip.
- **Authority Home**: Scope Selector, Executive Top Banner, 5 Metric cards, Cross-Mine Risk Index table, Urgent Action Queue, and Central Consolidated Governance Summary (`AuthorityDailyGovernanceSummary``).
- **Miner Home**: Worker Summary strip, Emergency SOS banner, 4 Quick Action cards, and My Recent Field Submissions table.

21. API / BACKEND / SERVICES

Endpoint	Method	Purpose	Request Data	Response Data	Real / Mocked
<code>/api/auth/login`</code>	<code>POST`</code>	User authentication	<code>{ email, password }`</code>	<code>{ token, user }`</code>	Real backend route
<code>/api/field-reports`</code>	<code>POST`</code>	Create field report	<code>{ category, description, severity, zone }`</code>	<code>{ report }`</code>	Real backend route
<code>/api/minesign/events`</code>	<code>POST`</code>	Ingest gesture candidate	<code>{ gesture, confidence, worker_id, zone }`</code>	<code>{ event_id, status }`</code>	Real backend route
<code>/api/minesign/events/:id/confirm`</code>	<code>PATCH`</code>	Confirm gesture candidate	<code>{ confirmed_by, confirmation_notes }`</code>	<code>{ event }`</code>	Real backend route
<code>/api/emergency/sos`</code>	<code>POST`</code>	Trigger SOS beacon	<code>{ latitude, longitude, emergency_type }`</code>	<code>{ sos_event }`</code>	Real backend route
<code>/api/models/predict-risk`</code>	<code>POST`</code>	Compute Model 4 risk	<code>{ equipment_age, methane_level, ... }`</code>	<code>{ predicted_risk_pct, ... }`</code>	Real backend route
<code>/api/models/send-assessment`</code>	<code>POST`</code>	Snapshot assessment to Auth	<code>{ model_id, inputs, output }`</code>	<code>{ assessment }`</code>	Real backend route
<code>/api/violations/:id/step2`</code>	<code>PATCH`</code>	Authority 2-step closure	<code>{ verification_notes }`</code>	<code>{ violation }`</code>	Real backend route
<code>/api/audit/verify-integrity`</code>	<code>POST`</code>	Verify SHA-256 chain	None	<code>{ valid, totalLogs }`</code>	Real backend route
<code>/api/workflow/generate-statutory-report/:mineId`</code>	<code>GET`</code>	Stream PDF report	Query params	Binary PDF Stream	Real backend route

22. ENVIRONMENT VARIABLES / CONFIGURATION

- `PORT``: Server port (Default: `5001``).
- `DATABASE_URL``: PostgreSQL / Supabase connection URI (If omitted, uses embedded `mockStore.js``).
- `JWT_SECRET``: Secret key for signing authentication tokens.
- `NODE_ENV``: Environment mode (`development`` / `production``).
- `VITE_API_URL``: Frontend API base URL.

23. AUTHENTICATION / USER ROLES

- **Roles**: `miner``, `supervisor``, `authority`` (with aliases `corporate``, `regulator``).
- **Quick Role Switcher**: Demo switcher calls `/api/auth/quick-login/:role`` to generate a valid JWT token for testing role perspective without re-entering credentials.

- **Session Handling**: Token stored in `localStorage.coalguard_token`` and passed via `Authorization: Bearer <token>` header.

24. ERROR / BUG / TODO INVENTORY

A. Confirmed Behaviors & Demo Design

- **Camera Feeds**: Client CCTV feeds display pre-recorded MP4 videos (`people_counter_demo.mp4``, `helmet_detection_demo.mp4``, `smoke_detection_demo.mp4``) with real interactive API buttons simulating detection triggers.
- **Python Inference Server**: Standalone Flask inference server (`minesign_ai/inference_server.py`` on port 5005) is auto-spawned by `server/routes/minesign.js``.

B. Potential Risks

- **Port 5005 Spawning**: If Python virtual environment (`minesign_ai.venv``) is not initialized, `ensureInferenceServer()` logs a fallback warning and uses API-based direct gesture transmission.
- **Local Storage Reliance**: Section read notifications rely on browser `localStorage``. Clearing browser data resets seen timestamps.

25. CURRENT FUNCTIONALITY STATUS

Portal	Feature	Status	Notes
Miner	Biometric Check-In	Fully Implemented	Calls <code>/api/attendance/check-in`</code>
Miner	Voice Hazard Report	Fully Implemented	Speech-to-Text + NLP classification API
Miner	MineSign Camera Gestures	Fully Implemented	Real webcam MediaPipe + direct gesture trigger fallback
Miner	Emergency SOS Beacon	Fully Implemented	Captures GPS coordinates & triggers high-priority alarm
Supervisor	CCTV People Counter	Implemented (Simulated Video)	Video feed + live counter sync & headcount broadcast API
Supervisor	CCTV Helmet Detection	Implemented (Simulated Video)	Video feed + Dismiss / Raise Field Issue API
Supervisor	CCTV Smoke & Fire Vision	Implemented (Simulated Video)	Video feed + Field Report / Authority Escalation API
Supervisor	MineSign Confirmation	Fully Implemented	<code>PENDING_CONFIRMATION`</code> candidate triage & SHA-256 log
Supervisor	Mine Risk Scoring (Model 4)	Fully Implemented	Real tabular calculation engine
Supervisor	What-If Simulator (Model 6)	Fully Implemented	Multi-factor risk slider simulation engine
Supervisor	Send Assessment to Authority	Fully Implemented	Deep-clone snapshot creation & dispatch
Supervisor	Daily Operational Log	Fully Implemented	Timeline rendering of operational events
Authority	2-Step Violation Closure	Fully Implemented	Step 1 photo proof -> Step 2 Authority physical sign-off
Authority	AI Assessments Received	Fully Implemented	Views snapshot, risk drivers, and updates review status
Authority	Escalated Emergencies Queue	Fully Implemented	Authority receives ONLY supervisor-escalated emergency events
Authority	Communication Approval	Fully Implemented	Review requests & direct statutory directive dispatch
Authority	SHA-256 Audit Trail	Fully Implemented	Immutable hash chain + tamper verification engine
Authority	PDF Report Generator	Fully Implemented	PDFKit document streaming
GIS Map	Sector GIS Polygons & Pins	Fully Implemented	Leaflet interactive map centered on Jharia Coalfield

26. IMPORTANT FILE MAP

- **Miner Portal**: `client/src/pages/miner/MinerDashboard.jsx``
- **Supervisor Portal**: `client/src/pages/supervisor/SupervisorDashboard.jsx``
- **Authority Portal**: `client/src/pages/authority/AuthorityDashboard.jsx``
- **MineSign AI Camera & View**: `client/src/components/minesign/MineSignView.jsx``, `client/src/components/minesign/MinerCameraModal.jsx``, `server/routes/minesign.js``, `minesign_ai/inference_server.py``
- **CCTV Surveillance**: `client/src/components/surveillance/CCTVSurveillanceView.jsx``
- **Risk Scoring & What-If**: `client/src/components/analytics/Model4RiskScoringView.jsx``, `client/src/components/analytics/Model6WhatIfSimulatorView.jsx``, `server/routes/mlModels.js``
- **Authority AI Assessments**: `client/src/components/authority/AuthorityAIAssessmentsView.jsx``
- **Two-Step Violation Modal**: `client/src/components/common/TwoStepViolationModal.jsx``, `server/routes/violations.js``
- **Cryptographic Audit Trail**: `server/routes/auditTrail.js``, `server/db/hashChain.js``
- **GIS Map**: `client/src/components/maps/MineGISMap.jsx``, `server/routes/gis.js``
- **Auth & State**: `client/src/context/AuthContext.jsx``, `server/routes/auth.js``

- **Backend Entry & DB**: `server/index.js`, `server/db/pool.js`, `server/db/mockStore.js`

27. FINAL SYSTEM ARCHITECTURE

[illegible]

28. MOST IMPORTANT FINAL SUMMARY

A. WHAT IS CURRENTLY WORKING

- Role-based login and quick role switcher ('Miner', 'Supervisor', 'Authority').
- Biometric attendance check-in and Form B muster sync.
- Voice hazard reporting with Web Speech API and NLP classification.
- MineSign gesture reporting with real MediaPipe webcam inference and direct button fallback.
- Gestural candidate confirmation workflow ('PENDING_CONFIRMATION' -> 'OPEN').
- Emergency SOS distress beacon broadcasting GPS coordinates.
- CCTV surveillance controls (People counter sync, Helmet issue creation, Smoke routing).
- Model 4 Risk Scoring and Model 6 What-If Simulator calculations.
- "Send Assessment to Authority" snapshot creation and dispatch.
- 2-Step Violation Closure workflow (Step 1 photo proof -> Step 2 Authority physical sign-off).
- Supervisor '!' Authority communication request approval workflow and direct directive dispatch.
- Cryptographic SHA-256 audit trail with live integrity verification and tamper simulation.
- PDF statutory report generation via PDFKit.
- Leaflet GIS spatial map with zone polygons and GPS pins.

B. WHAT IS CURRENTLY IMPLEMENTED

- Complete 3-portal UI codebase (`MinerDashboard`, `SupervisorDashboard`, `AuthorityDashboard`).
- Complete backend REST API suite (`18 Express route modules`).
- Dual-tier data engine (PostgreSQL pool with embedded `mockStore` fallback).
- Python AI model scripts (`minesign\_ai`, `helmet detection`, `people counter`, `SMOKE DETECTION`, `model 4`, `model 5`, `model 6`).

C. WHAT IS SIMULATED/MOCKED

- CCTV feeds use pre-recorded MP4 demonstration videos (`people\_counter\_demo.mp4`, `helmet\_detection\_demo.mp4`, `smoke\_detection\_demo.mp4`) with live functional API integration buttons.
- Biometric scanner hardware uses client-side confirmation triggers simulating DGMS Form B muster roll entries.

D. WHAT IS PARTIALLY IMPLEMENTED

- Gas Leak Risk Model UI placeholder card reserved under AI Models tab for future IoT sensor stream integration.

E. CONFIRMED ISSUES

- No breaking errors detected. The system functions cleanly across all portals in demo mode.

F. POTENTIAL RISKS

- Python virtual environment (`~minesign_ai/.venv`) must exist for live webcam MediaPipe gesture inference on port 5005; if unavailable, the system falls back to direct gesture transmission buttons.

G. IMPORTANT FILES

- ``client/src/App.jsx``
- ``client/src/context/AuthContext.jsx``
- ``client/src/pages/miner/MinerDashboard.jsx``
- ``client/src/pages/supervisor/SupervisorDashboard.jsx``
- ``client/src/pages/authority/AuthorityDashboard.jsx``
- ``client/src/components/authority/AuthorityAIAssessmentsView.jsx``
- ``client/src/components/common/TwoStepViolationModal.jsx``
- ``server/index.js``
- ``server/db/mockStore.js``
- ``server/routes/mlModels.js``
- ``server/routes/minesign.js``
- ``server/routes/communication.js``

H. IMPORTANT ROUTES

- Frontend: State-based tab routes (``home``, ``gestures``, ``my_reports``, ``ask_supervisor``, ``overview``, ``cameras``, ``models``, ``emergencies``, ``field_reports``, ``compliance``, ``map``, ``communication``, ``ai_assessments``, ``audit``).
- Backend APIs: ``/api/auth``, ``/api/field-reports``, ``/api/minesign/events``, ``/api/emergency/sos``, ``/api/models/predict-risk``, ``/api/models/simulate-whatif``, ``/api/models/send-assessment``, ``/api/violations/:id/step2``, ``/api/communication/requests``, ``/api/audit/verify-integrity``.

I. IMPORTANT DATA FLOWS

- Miner SOS / Hazard Report -> Supervisor Field Reports -> Authority Escalation.
- Miner Gestures -> MineSign AI Candidate (``PENDING_CONFIRMATION``) -> Supervisor Confirmation (``OPEN``) -> SHA-256 Audit Log.
- Supervisor What-If Simulator -> "Send Assessment" -> Immutable Snapshot -> Authority Review (``NEW`` -> ``REVIEWED``).
- Violation -> Step 1 Photo Proof -> Step 2 Authority Physical Sign-off (``verified_closed``).

J. IMPORTANT ENVIRONMENT VARIABLES

- ``PORT`` (default ``5001``)
- ``DATABASE_URL`` (optional Postgres connection string)
- ``JWT_SECRET``
- ``NODE_ENV``

K. FEATURES THAT MUST NOT BE BROKEN WHEN MODIFYING THE PROJECT

- Role separation logic in ``App.jsx``.
- SHA-256 cryptographic audit chain logging in ``server/db/mockStore.js`` and ``server/routes/auditTrail.js``.
- Candidate confirmation state machine for MineSign gesture events (``PENDING_CONFIRMATION`` -> ``OPEN`` / ``REJECTED``).
- Two-Step Violation Closure state machine (``open`` -> ``rectification_submitted`` -> ``verified_closed``).
- Immutability of transmitted AI Assessment snapshots in ``mockStore.ai_assessments``.
- Dual fallback database architecture (``PostgreSQL`` / ``mockStore.js``).

